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Department of Employment and Training

Course : TNPSC Group I Mains Material

Subject : Role and impact of Science and Technology in the development of India

Topic : Advancements in the fields of Remote Sensing and its benefits

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REMOTE SENSING

The observation for the Earth from space is usually termed as remote sensing and satellites used for the purpose are known as Remote Sensing Satellites. The Remote Sensing Satellites (RSS) is a 'Space-based' Earth observation system, it is a technique used for collecting information of any object on Earth from about 900 km above the surface of the Earth, through the measurement of Electro Magnetic Spectrum of the Sun that is reflected, scattered and remitted by objects on Earth's surface. Indian Earth Observation System operates in Low Earth Orbit (LEO), Medium Earth Orbit (MEO) and Geostationary Earth Orbit (GEO).

Most of the RSS are polar, sun-synchronous satellites in low Earth orbits (LEO) that is their latitudinal position above the Earth is 1000, their movement is in pole to pole direction & their angular appearance in relation to the Sun is fixed. Such an orbit is chosen because it enables the satellite to take Earth's pictures always at a fixed angle of illumination by the Sun.

The RSS has the ability to provide a synoptic, view of large areas for systematic and speedy collection & distribution of metrology, geology, geography, agriculture date within a short time period. Thus, the RSS provides key tips for planning & forms an integral part of resource management for the exploitation of renewable & natural resources of the Earth.

Starting with IRS-1A in 1988, ISRO has launched many operational remote sensing satellites. The data from these satellites are used for several applications covering agriculture, water resources, urban planning, rural development mineral prospecting, environment, forestry, ocean resources and disaster management. Today, India has one of the largest constellations of remote sensing satellites in operation. Currently, more than ten operational satellites are in Sun-synchronous orbit – RESOURCES AT- 1,2, 2A CARTOSAT – 1, 2 2A, 2B, RISAT-1 and 2. OCEANSAT – 2 , Megha Tropiques, SARAL,

SCATSAT-1, and Hysis and 'four' in Geostationary orbit – INSAT – 3D, Kalpana & INSAT 3A, INSAT-3DR, Hysis being the latest.

Applications of Remote Sensing Satellites:

The space borne remote sensing via satellites constitute a far greater improvement over-groundbased 'or' even air-borne surveys because:

1. It is cheaper & much speedier
2. It is more accurate & reliable
3. It is a multi-dimensional survey tool whereby more than one type of surveys can be simultaneously carried out for a wide variety of application.

The remote sensing in India is an umbrella of the National Natural Resources Management System (NNRMS), Remote sensing has been applied in many diverse fields in India which are:

1. **Agriculture:** In the field of agriculture, the RSS has helped in assessing the net crop area, Crop production forecasting, Movement of locust, Flood warming, damage assessments. On the basics of data provided by 'IRS-1A 1B', Planning Commission has divided the country into 15 Agro-climatic regions for better agriculture production.
2. **Natural Resource Management:** In the field of natural resource management, it has helped in accurately assessing the Land use and land cover mapping, mineral prospecting, forest resource surveys, Survey of forest cover distribution, monitoring of forest-fires etc. National Westerland Maps have been prepared, identifying the types and extent of wastelands at the village level, for 237 selected districts spread all over the country. These maps have been provided to the Department of Wasteland Development and district authorities for taking up Wasteland reclamation measures.
3. **Disaster Management & Meteorology:** The RSS finds its applications in the early warning mechanism for cyclones, the short-term weather forecasting, and the warming, Monitoring as well as the impact assessment for droughts and floods.

4. **Sustainable Development:** A major application of IRS data is the Integrated Mission for Sustainable Development (IMSD) which was initiated in 1992 to generate locale-specific action plans for sustainable development of natural resources, in terms of productivity, quality of the environment and ecological set-up. The locale-specific prescriptions are generated under IMSD using data Iron IRS satellites and collateral socio-economic data in collaboration with the State Remote Sensing Application Centre, Universities, private entrepreneurs and non-governmental organizations.
5. **Urban Studies:** With the availability of high-resolution panchromatic data from IRS-IC and IRS-ID, many new application in urban studies have been carried out such as urban sprawl, structural plans for cities, guide maps for Delhi, Bangladesh etc.
6. **Defence:** The equipment on board IRS systems of satellites can act as a spy in the sky. They can pick up images of troop movements as well as deployments by the enemy. Remote Sensing has also helped to track a lost River bed in the Thar Desert of the country which could be used for extending drinking H_2O facility to the population.

Some Notes on Above Satellites

1. HySIS Hyper Spectral Imaging Satellite is an Earth observation satellite which carried a payload named Hyperspectral Imager in VNIR and SWIR bands build around ISRO's Mini Satellite-2 (IMS-2) bus. The primary goal of HySIS is to study the Earth's surface in visible, near infrared and shortwave infrared regions of the electromagnetic spectrum.
2. It will provide a repetitive basis to users and supplementing the data from the existing Malti spectral sensors. A wide range of applications in agriculture, forestry, geological environment, coastal zones and inland water etc., are derived from the satellite.
3. CARTOSAT is an Earth observation satellite that carries a state-of-the-art panchromatic (PAN) camera that provides scene-specific spot imagery. The data from the satellite will be used for cartographic applications like mapping, urban and rural infrastructure development and management as well as application in

Land Information System (LIS) and Geographical Information System (GIS), and DEM generation. The Cartosat-2 series is a dedicated satellite for the defence forces to strengthen the defence military apparatus of the country.

Significance of Cartosat

1. Cartosat-2 is a remote-sensing satellite. Its objective is to provide high-resolution scene-specific spot imagery. The images sent by the satellite would be useful for cartographic applications, urban, rural and coastal land use and regulation, utility management like road network monitoring, water distribution, the creation of land use maps, and various other land Information Systems (LIS) as well as Geographical Information Systems (GIS).
2. The satellite can provide India's defence forces scene-specific imagery in the military's area of interest. Thus, India can track developments along the borders with China and Pakistan. It can be used to identify terrorist camps and bunkers and some enemy formations.
3. However, the Indian military wants more dedicated satellites for exclusive military use. So far only the Indian Navy has a dedicated satellite. India has moved towards network-centric warfare in which land, air and sea defence forces are networked through space technology and surveillance aircraft.

This satellite series is designed and developed for the defence forces to create a robust mechanism with their infrastructure, which includes ground stations and trained mans power to access data. Defence surveillance will get enhanced because it can be used in identifying terrorist camps and bunders

1. **OCEANSATs** are used in Ocean and atmospheric studies (like atmospheric humidity and temperature vertical profiles from ROSA).
2. **RESOURSESAT** is used in Resource Monitoring applications. Resources at 2A will give regular micro and macro information on land and water bodies below, farm lands and crop extent, forests, mineral deposits, coastal information, rural and urban spreads besides helping in disaster management.
3. **Megha-Tropiques** is used in Meteorological and Oceanographic applications. RISAT is used for Agriculture mapping, disaster management support activities.

4. **SARAL** or satellite with ARogs and ALtiKa is a cooperative altimetry technology mission of ISRO and CNES (Space Agency of France). SARAL will aid work in many fields like Climate monitoring; Ocean, Earth system and climate research; Continental ice studies; Protection of biodiversity; Improvement of maritime security etc.
5. Current, INSAT-3A, KALPANA-1, INSAT-3DR are providing meteorological data to the user community. INSAT-3DR is an advanced weather satellite put in the Geosynchronous Transfer Orbit (GTO) and then transferred into the Geosynchronous Transfer Orbit (GTO). The INSAT-3DR will generate images of the Earth form a geostationary attitude of 36,000 km every 26 minutes and provide information on parameters such as sea surface temperature, snow cover, could motion winds, among others. The major users of the service will be the Indian Coast Guards, Airports Authority of India (AAI), Directorate General of Shipping, Defence Services and fishermen.
6. **RADAR IMAGING SATELLITES (RISAT):** It has an all-weather-imaging sensor capable of taking images in cloudy and snow-covered regions and also both during day and night. ISRO currently operates two Radar Imaging Satellites – RISAT-1 and RISAT-2.
7. **SCATSAT-1** is a continuity mission for Oceansat-2 Scatterometer to provide wind vector data products for weather forecasting, cyclone detection and tracing services.

Future IRS Launches

Following are the remote sensing satellites planned by ISRO to be launched next strengthening the fleet of IRS satellites and widening their applications:

1. **RESOURCESAT-3 –** A follow on to Resourcesat-2, it will carry more advanced LISS-III-WS (Wide Swath) Sensor having similar swath and revisit capability as Advanced Wide Field Sensor (AWiFS) thus overcoming any spatial resolution limitation of AWiFS. The satellite would also carry Atmospheric Correction Sensor (ACS) for quantitative interpretation and geophysical parameter retrieval.

2. **CARTOSAT-3:** A continuation of Cartosat series, it will have a resolution 30 cm and 6 km swath suitable for cadastre and infrastructure mapping and analysis. It would also enhance disaster monitoring and damage assessment.
3. **OCEANSAT-3**: Oceansat-3 would carry Thermal IR Sensor, 12 channel Ocean Colour Monitor, Scatterometer and Passive Microwave Radiometer. IR Sensor and Ocean Colour Monitor would be used in the analysis for operational Potential Fishing Zones. Satellite is mainly for Ocean biology and sea state applications.
4. **VILLAGE RESOURCE CENTRE (VRC):** Combining the services offered by INSAT and IPS satellites, a new concept namely Village Resource Centre (VRC) is envisaged to provide information on natural resources, land and water resources management, telemedicine, tele-education, adult education, vocational training, health and family welfare programmes have been established. Nearly 500 such VRCs have been established in the country.

Questions:

1. Describe the applications of Remote Sensing Satellites.
2. Write a short note on CARTOSAT and its significance.
3. Mention some of the important Future IRS Launches. Discuss.